

Geometry (8 th std)

Choose the Appropriate Answer

1. Two similar triangles will always have _____ angles.
(a) Acute (b) Obtuse (c) Right (d) Matching

Ans: (d) Matching

2. If in triangles PQR and XYZ, $\angle P = \angle X$ and $\angle Q = \angle Y$ then they will be similar if
(a) $\angle Q = \angle Y$ (b) $\angle P = \angle X$ (c) $\angle Q = \angle X$ (d) $\angle P \cong \angle Z$

Ans: (c) $\angle Q = \angle X$

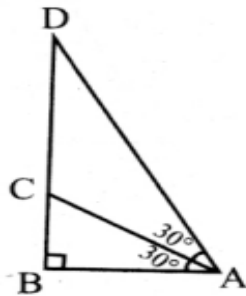
3. A flag pole 15 m high casts a shadow of 3m at 10 (a)m. The shadow cast by a building at the same time is 18.6m. The height of the building is
(a) 90 m (b) 91 m (c) 92 m (d) 93 m

Ans: (d) 93 m

4. If $\triangle ABC \sim \triangle PQR$ in which $\angle A = 53^\circ$ and $\angle Q = 77^\circ$, then $\angle R$ is
(a) 50° (b) 60° (c) 70° (d) 80°

Ans: (a) 50°

5. In the figure, which of the following statements is true?



- (a) $AB = BD$ (b) $BD < CD$ (c) $AC = CD$ (d) $BC = CD$

Ans: (c) $AC = CD$

6. Corresponding sides of similar triangles are _____.
(a) Similar (b) Congruent (c) Proportional (d) None of these

Ans: (c) Proportional

7. Similar triangles have the same _____ but not necessarily the same size.
(a) Sides (b) Angles (c) Shape (d) None of these

Ans: (c) Shape

8. In any triangle _____ sides are opposite to equal angles.
(a) Not Equal (b) Equal (c) Opposite (d) Same

Ans: (b) Equal

9. The symbol ' $\cong$ ' is used to represent _____ triangles.
(a) Similar (b) Congruent (c) Proportional (d) None of these

Ans: (b) Congruent

10. The symbol $\sim$ is used to represent _____ triangles.
 (a) Similar (b) Congruent (c) Proportional (d) None of these

Ans: (a) Similar

11. If in a ΔPQR , $PR^2 = PQ^2 + QR^2$, then the right angle of ΔPQR is at the vertex _____.

(a) P (b) Q (c) R (d) None of these

Ans: (b) Q

12. If 'l' and 'm' are the legs and 'n' is the hypotenuse of a right angled triangle then, $l^2 =$ _____.

(a) $n^2 - m^2$ (b) $m^2 - n^2$ (c) $m^2 + n^2$ (d) None of these

Ans: (a) $n^2 - m^2$

13. If the sides of a triangle are in the ratio 5:12:13 then, it is a _____.

(a) Acute angle triangle (b) Obtuse angle triangle
 (c) Right angled triangle (d) None of these

Ans: (c) Right angled triangle

14. The medians of a triangle cross each other at _____.

(a) Altitude (b) Bisector (c) Incentre (d) Centroid

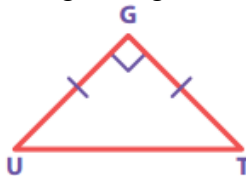
Ans: (d) Centroid

15. The centroid of a triangle divides each medians in the ratio _____.

(a) 1: 2 (b) 2: 1 (c) 1: 3 (d) 3: 1

Ans: (b) 2: 1

16. If ΔGUT is isosceles and right angled, then $\angle TUG$ is _____.



(a) 30° (b) 40° (c) 45° (d) 55°

Ans: (c) 45°

17. The hypotenuse of a right angled triangle of sides 12cm and 16cm is _____.

(a) 28 cm (b) 20 cm (c) 24 cm (d) 21 cm

Ans: (b) 20 cm

$$c^2 = a^2 + b^2 = 12^2 + 16^2 = 144 + 256 \Rightarrow c^2 = 400 \Rightarrow c = 20 \text{ cm}$$

18. The area of a rectangle of length 21cm and diagonal 29 cm is _____.

(a) 609 cm^2 (b) 580 cm^2 (c) 420 cm^2 (d) 210 cm^2

Ans: (c) 420 cm^2

$$21^2 + x^2 = 29^2 \Rightarrow x^2 = 841 - 441 = 400 \Rightarrow x = 20$$

$$\text{Area of the rectangle} = \text{length} \times \text{breadth} = 21 \text{ cm} \times 20 \text{ cm} = 420 \text{ cm}^2$$

19. The sides of a right angled triangle are in the ratio 5:12:13 and its perimeter is 120 units then, the sides are.

- (a) 25, 36, 59 (b) 10, 24, 26 (c) 36, 39, 45 (d) 20, 48, 52

Ans: (d) 20, 48, 52

Its perimeter is $5a + 12a + 13a = 30a = 120$ units $\Rightarrow a = 4$ units

20. The sum of the three angles of a triangle is _____.

- (a) 90° (b) 270° (c) 180° (d) 360°

Ans: (c) 180°

21. The exterior angle of a triangle is equal to the sum of the _____ angles to it.

- (a) Vertically opposite (b) Interior (c) Congruent (d) None of these

Ans: (b) Interior

22. In a triangle, the sum of any two sides is _____ than the third side.

- (a) Greater (b) Less (c) Equal (d) None of these

Ans: (a) Greater

23. Angles opposite to equal sides are _____ and vice – versa (a)

- (a) Congruent (b) Similar (c) Equal (d) None of these

Ans: (c) Equal

24. The area of the trapezium is _____ sq. units.

- (a) $\frac{1}{2} \times b \times h$ (b) $\frac{1}{2} \times h \times (a + b)$ (c) $\frac{1}{2} \times (a + b)$ (d) None

Ans: (b) $\frac{1}{2} \times h \times (a + b)$

25. The distance between the parallel sides of a trapezium is called as its _____.

- (a) Base (b) Length (c) Height (d) Breadth

Ans: (c) Height

26. If the height and parallel sides of a trapezium are 5cm, 7cm and 5cm respectively, then its area is _____ sq. cm.

- (a) 24 (b) 28 (c) 32 (d) 30

Ans: (d) 30

Area of trapezium = $\frac{1}{2} \times h \times (a + b)$ sq. units = $\frac{1}{2} \times 5 \times (7 + 5) = 30$ sq. cm

27. In an isosceles trapezium, the non-parallel sides are _____ in length.

- (a) Equal (b) Not equal (c) Greater (d) Lesser

Ans: (a) Equal

28. To construct a trapezium, _____ measurements are enough.

- (a) 2 (b) 3 (c) 4 (d) 6

Ans: (c) 4

29. If the area and sum of the parallel sides are 60 cm^2 and 12 cm, its height is _____.

- (a) 20 cm (b) 15 cm (c) 10 cm (d) 5 cm

Ans: (c) 10 cm

$$\text{Area of the trapezium} = \frac{1}{2} \times h (a + b)$$

$$60 = \frac{1}{2} \times h \times (12) \Rightarrow h = (60 \times 2) / 12 = 10 \text{ cm}$$

30. What is the appropriate condition to construct a quadrilateral?

- (a) When four sides and one diagonal are given
- (b) When three sides and one diagonal are given
- (c) When two sides and one diagonal are given
- (d) None of the above

Ans: (a) When four sides and one diagonal are given

31. If two diagonals and three sides are given, then:

- (a) A quadrilateral cannot be constructed
- (b) A quadrilateral can be constructed
- (c) Insufficient information
- (d) Any polygon can be constructed

Ans: (b) A quadrilateral can be constructed

32. To construct a quadrilateral, we need to know two adjacent side and _____ angles.

- (a) One
- (b) Two
- (c) Three
- (d) All four angles

Ans: (c) Three

33. To construct a quadrilateral, we need to know two diagonals and _____ sides.

- (a) One
- (b) Two
- (c) Three
- (d) All four sides

Ans: (c) Three

34. To construct a quadrilateral, we need to know three sides and _____ included angles.

- (a) One
- (b) Two
- (c) Three
- (d) All four angles

Ans: (b) Two

35. To construct a square, we need to know:

- (a) All the interior angles
- (b) All the side lengths
- (c) Only one interior angle
- (d) Only one side length

Ans: (d) Only one side length

36. To construct a rectangle, we need to know:

- (a) All the interior angles
- (b) All the Sides
- (c) Only Length and breadth
- (d) Only one angle measure

Ans: (c) Only Length and breadth

37. If two diagonals are given, then we can construct a:

- (a) Rhombus
- (b) Rectangle
- (c) Kite
- (d) Parallelogram

Ans: (a) Rhombus

38. How many measurements are required to construct a quadrilateral, uniquely?

- (a) Four
- (b) Five
- (c) Six
- (d) Three

Ans: (b) Five

39. To construct a parallelogram we need to know:

- (a) Length of its parallel sides (b) Measure of interior angles
(c) Two adjacent sides and one angle (d) Two adjacent sides and two angles

Ans: (c) Two adjacent sides and one angle

40. A polygon that has a minimum number of sides is:

- (a) Triangle (b) Square (c) Rectangle (d) Angle

Ans: (a) Triangle

41. If n is the number of sides, then the number of diagonals of a polygon is:

- (a) $n / 2$ (b) $n / 3$ (c) $n(n - 3) / 2$ (d) $n(n - 3) / 3$

Ans: (c) $n(n - 3) / 2$

42. The sum of all the interior angles of a hexagon is:

- (a) 720° (b) 540° (c) 360° (d) 180°

Ans: (a) 720°

43. If the sum of interior angles of a regular polygon is 540° . Find the name of the polygon.

- (a) Quadrilateral (b) Pentagon (c) Hexagon (d) Heptagon

Ans: (b) Pentagon

44. _____ measurements can determine a quadrilateral uniquely.

- (a) Three (b) Four (c) Five (d) Six

Ans: (c) Five

45. The sum of exterior angles of a polygon is equal to:

- (a) 180° (b) 360° (c) 540° (d) 720°

Ans: (b) 360°

46. All the sides of a regular polygon are:

- (a) Equal in length (b) Unequal in length
(c) Parallel to each other (d) None of these

Ans: (a) Equal in length

47. If a polygon has 8 sides, then the number of diagonals it has is:

- (a) 8 (b) 16 (c) 20 (d) 24

Ans: (c) 20

48. Can we draw a square with a side length equal to 7cm?

- (a) Yes (b) No (c) Cannot be determined (d) None of the above

Ans: (a) Yes

49. If a quadrilateral constructed has two distinct consecutive sides equal in length, then it is a:

- (a) Rhombus (b) Kite (c) Parallelogram (d) Rectangle

Ans: (b) Kite

50. Which of the following is not a quadrilateral?

- (a) Square (b) Rectangle (c) Triangle (d) Parallelogram

Ans: (c) Triangle

51. Which of the following quadrilaterals has two pairs of adjacent sides equal and its diagonals intersect at 90 degrees?

- (a) Square (b) Kite (c) Rhombus (d) Rectangle

Ans: (b) Kite

52. Which one of the following is a regular quadrilateral?

- (a) Square (b) Trapezium (c) Kite (d) Rectangle

Ans: (a) Square

53. If AB and CD are two parallel sides of a parallelogram, then:

- (a) $AB > CD$ (b) $AB < CD$ (c) $AB = CD$ (d) None of the above

Ans: (c) $AB = CD$

54. The perimeter of a parallelogram whose parallel sides have lengths equal to 12 cm and 7 cm is:

- (a) 21 cm (b) 42 cm (c) 19 cm (d) 38 cm

Ans: (d) 38 cm

55. If $\angle A$ and $\angle C$ are two opposite angles of a parallelogram, then:

- (a) $\angle A > \angle C$ (b) $\angle A = \angle C$ (c) $\angle A < \angle C$ (d) None of the above

Ans: (b) $\angle A = \angle C$

56. If $\angle A$ and $\angle B$ are two adjacent angles of a parallelogram. If $\angle A = 70^\circ$, then $\angle B = ?$

- (a) 70° (b) 90° (c) 110° (d) 180°

Ans: (c) 110°

57. ABCD is a rectangle and AC & BD are its diagonals. If AC = 10 cm, then BD is:

- (a) 10 cm (b) 5 cm (c) 15 cm (d) 20 cm

Ans: (a) 10 cm

58. Each of the angles of a square is:

- (a) Acute angle (b) Right angle (c) Obtuse angle (d) 180 degrees

Ans: (b) Obtuse angle

59. The quadrilateral whose diagonals are perpendicular to each other is:

- (a) Parallelogram (b) Rectangle (c) Trapezium (d) Rhombus

Ans: (d) Rhombus

60. Which of the following is not a regular polygon?

- (a) Square (b) Equilateral triangle (c) Rectangle (d) Regular hexagon

Ans: (c) Rectangle

61. If the two angles of a triangle are 80° and 50° , respectively. Find the measure of the third angle.

- (a) 50° (b) 60° (c) 70° (d) 80°

Ans: (a) 50°

62. In a parallelogram ABCD, angle A and angle B are in the ratio 1:2. Find the angle

(a)

(a) 30°

(b) 45°

(c) 60°

(d) 90°

Ans: (c) 60°

63. The angles of a quadrilateral are in ratio 1:2:3:4. Which angle has the largest measure?

(a) 120°

(b) 144°

(c) 98°

(d) 36°

Ans: (b) 144°

64. The length and breadth of a rectangle is 4 cm and 2 cm respectively. Find the perimeter of the rectangle.

(a) 12 cm

(b) 6 cm

(c) 8 cm

(d) 16 cm

Ans: (a) 12 cm

65. The diagonals of a rectangle are $2x + 1$ and $3x - 1$, respectively. Find the value x.

(a) 1

(b) 2

(c) 3

(d) 4

Ans: (b) 2

66. The diagonals of a kite:

(a) Bisects each other

(b) Are perpendicular to each other

(c) Does not bisect each other

(d) None of the above

Ans: (b) Are perpendicular to each other

67. A rhombus has a side length equal to 5 cm. Find its perimeter.

(a) 25

(b) 10

(c) 20

(d) 30

Ans: (c) 20

68. ABCD is a parallelogram. If angle A is equal to 45° , then find the measure of its adjacent angle.

(a) 135°

(b) 120°

(c) 115°

(d) 180°

Ans: (a) 135°

69. Which of the following quadrilaterals has two pairs of adjacent sides equal and diagonals intersecting at right angles?

(a) Square

(b) Rhombus

(c) Kite

(d) Rectangle.

Ans: (c) Kite

70. Which of the following quadrilaterals has a pair of opposite sides parallel?

(a) Rhombus

(b) Trapezium

(c) Kite

(d) Rectangle.

Ans: (b) Trapezium

71. Which of the following quadrilaterals is a regular quadrilateral?

(a) Rectangle

(b) Square

(c) Rhombus

(d) Kite.

Ans: (b) Square

72. Which of the quadrilaterals has all angles as right angles, opposite sides equal and diagonals bisect each other?

(a) Rectangle

(b) Rhombus

(c) Square

(d) None of these.

Ans: (a) Rectangle

73. Which of the parallelograms has all sides equal and diagonals bisect each other at right angle?

- (a) Square (b) Rectangle (c) Rhombus (d) Trapezium

Ans: (c) Rhombus

74. In an isosceles parallelogram, we have:

- (a) Pair of parallel sides as equal (b) Pair of non-parallel sides as equal
(c) Pair of non-parallel sides as perpendicular (d) None of these.

Ans: (b) Pair of non-parallel sides as equal

75. Which of the following is true for the adjacent angles of a parallelogram?

- (a) They are equal to each other (b) They are complementary angles
(c) They are supplementary angles (d) None of these.

Ans: (c) They are supplementary angles

76. The sides of a pentagon are produced in order. Which of the following is the sum of its exterior angles?

- (a) 540° (b) 180° (c) 720° (d) 360°

Ans: (d) 360°

77. Which of the following is a formula to find the sum of interior angles of a quadrilaterals of n-sides?

- (a) $\frac{n}{2} \times 180^\circ$ (b) $\left(\frac{n+1}{2}\right) \times 180^\circ$ (c) $\left(\frac{n-1}{2}\right) \times 180^\circ$ (d) $(n-2) \times 180^\circ$

Ans: (d) $(n-2) \times 180^\circ$

78. Diagonals of which of the following quadrilaterals do not bisect it into two congruent triangles?

- (a) Rhombus (b) Trapezium (c) Square (d) Rectangle.

Ans: (b) Trapezium

79. Which of the following is the once of a rhombus?

- (a) Product of its diagonals (b) $1/2$ (sum of its diagonals)
(c) 2 (Product of its diagonals) (d) 2 (Product of its diagonals)

Ans: (c) 2 (Product of its diagonals)

80. By which congruence rule are the following triangles congruent?

- (a) RHS (b) ASS (c) SSS (d) None of these

Ans: (a) RHS

81. How many pairs of adjacent angles are formed when two lines intersect at a point?

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (c) 4

82. The length of the perpendicular drawn from the opposite vertex to any side is:

- (a) Altitude (b) Median (c) Perpendicular bisector (d) Angle bisector

Ans: (a) Altitude

83. It is given that $\triangle ABC \cong \triangle FDE$. If $AB = 5\text{cm}$, $\angle B = 40^\circ$ and $\angle A = 80^\circ$, then the length of FD is:

- (a) 6 cm (b) 4cm (c) 5cm (d) None of these

Ans: (c) 5cm

84. Two distinct lines cannot have more than _____ point in common.

- (a) 1 (b) 2 (c) 3 (d) infinite

Ans: (a) 1

85. Two angles are congruent if they have

- (a) Same name (b) Unequal measures
(c) Equal measures (d) None of these

Ans: (c) Equal measures

86. What is the side included between the angles A and B in $\triangle ABC$?

- (a) AC (b) BC (c) AB (d) None of these

Ans: (c) AB

87. $\triangle ABC$ and $\triangle PQR$ are congruent under the correspondence: $ABC \leftrightarrow RQP$, then the part of $\triangle ABC$ that correspond to $\angle P$ is

- (a) $\angle A$ (b) $\angle C$ (c) $\angle B$ (d) None of these

Ans: (b) $\angle C$

88. If the vertical angle of an isosceles triangle is 40° , then measure of other two angles will be

- (a) $60^\circ, 60^\circ$ (b) $80^\circ, 80^\circ$ (c) $70^\circ, 70^\circ$ (d) $45^\circ, 45^\circ$

Ans: (c) $70^\circ, 70^\circ$

89. Two triangles, $\triangle PQR$ and $\triangle DEF$ are of the same size and shape. What can we conclude about them?

- (a) $\triangle PQR$ is smaller than $\triangle DEF$ (b) $\triangle PQR$ is larger than $\triangle DEF$
(c) $\triangle PQR$ is congruent to $\triangle DEF$ (d) $\triangle PQR$ is not congruent to $\triangle DEF$

Ans: (c) $\triangle PQR$ is congruent to $\triangle DEF$

90. In $\triangle ABC$ and $\triangle PQR$, $AB = 4\text{ cm}$, $BC = 5\text{ cm}$, $AC = 6\text{ cm}$ and $PQ = 4\text{ cm}$, $QR = 5\text{ cm}$, $PR = 6\text{ cm}$. then which of the following is true?

- (a) $\triangle ABC \cong \triangle QRP$ (b) $\triangle ABC \cong \triangle PQR$
(c) $\triangle ABC \cong \triangle RQP$ (d) None of these

Ans: (b) $\triangle ABC \cong \triangle PQR$

91. In $\triangle ABC$ and $\triangle DEF$, $AC = DF$, $AB = DE$ and $BC = EF$. By which property are $\triangle ABC$ and $\triangle DEF$ congruent?

- (a) R.H.S. property (b) S.S.S. property
(c) S.A.S. property (d) A.S.A. property

Ans: (b) S.S.S. property

92. A triangle in which all three sides are of equal lengths is called _____.

- (a) Isosceles (b) Equilateral (c) Scalene (d) None of these

Ans: (b) Equilateral

93. Which angle is included between the sides DE and EF of $\triangle DEF$?

- (a) $\angle F$ (b) $\angle D$ (c) $\angle E$ (d) None of these

Ans: (c) $\angle E$

94. In the quadrilateral ABCD, $AC = AD$ and AB bisect $\angle A$ and $\triangle ABC \cong \triangle ABD$. The relation between BC and BD is

- (a) $BC < BD$ (b) $BC > BD$ (c) $BC = BD$ (d) None of these

Ans: (c) $BC = BD$

95. If $\triangle DEF \cong \triangle BCA$, then the part of $\triangle BCA$ that correspond to $\angle E$ is

- (a) $\angle B$ (b) $\angle C$ (c) $\angle A$ (d) None of these

Ans: (b) $\angle C$

96. Two students drew a line segment each. What is the condition for them to be congruent?

- (a) They should be drawn with a scale
(b) They should be drawn on the same sheet of paper
(c) They should have different lengths
(d) They should have the same length.

Ans: (d) They should have the same length.

97. The measure of each angle of an equilateral triangle is:

- (a) 50° (b) 70° (c) 60° (d) 100°

Ans: (c) 60°

98. $\triangle ABC$ is right triangle in which $\angle A = 90^\circ$ and $AB = AC$. The values of $\angle B$ and $\angle C$ will be

- (a) $\angle B = \angle C = 30^\circ$ (b) $\angle B = \angle C = 50^\circ$ (c) $\angle B = \angle C = 45^\circ$ (d) $\angle B = \angle C = 60^\circ$

Ans: (c) $\angle B = \angle C = 45^\circ$

99. $\triangle ABC$ and $\triangle PQR$ are congruent under the correspondence: $ABC \leftrightarrow RPQ$, then the part of $\triangle ABC$ that correspond to PQ is

- (a) AC (b) AB (c) BC (d) None of These

Ans: (c) BC

100. What is the angle included between the sides PN and PM of $\triangle MNP$?

- (a) $\angle M$ (b) $\angle N$ (c) $\angle P$ (d) None of these

Ans: (c) $\angle P$